Python Packaging Cheat Sheet

uv is a fast Python package and project manager written in Rust.
Create a project with ’uv init’, add a dependency with ’uv add requests’,
and run a script inside the managed virtual environment with ’uv run main.py’.
The lock file uv.lock pins exact versions so installs are reproducible.

Testing with pytest

pytest discovers files named test_*.py and functions named test_*.
Use fixtures to share setup code, and parametrize to run one test with many inputs.
The flag -k selects tests by keyword and -x stops after the first failure.
For async code install pytest-asyncio and set asyncio_mode = auto.

Code quality tools

ruff is a linter and formatter that replaces flake8, isort and black.
mypy checks static type hints; run it in strict mode for new projects.
Enforcing PEP 8 automatically keeps pull requests free of style discussions.